PWPI

Multimodal Biosignal and Vision-Language Framework for Human Intention & Gesture Recognition

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Problem Statement

Current biosignal-based gesture/intent recognition systems face challenges due to:

- **Involuntary muscle contractions** causing false-positive EMG predictions.
- Limited availability of **domain-specific vision datasets.**
- Difficulty in reliably determining **user intention from a single modality (only EMG).**

Project Objective

> **Develop a multimodal framework that combines EEG, EMG, visual information and language-based reasoning to improve the reliability of human intention and gesture recognition.**

Literature - Starting Point

The project was initiated from the research paper:

IEEE Sensors Council IEEE SENSORS JOURNAL, VOL. 25, NO. 12, 15 JUNE 2025 22677

Lightweight Vision Language Model-Guided Gesture Recognition Based on Electromyography

Mustapha Deji Dere, Saehyung Cheong, Ji-Hun Jo, Giwon Ku, and Boreom Lee, *Member, IEEE*

The paper proposed following pipeline (EMG + Vision based with fixed prompts)

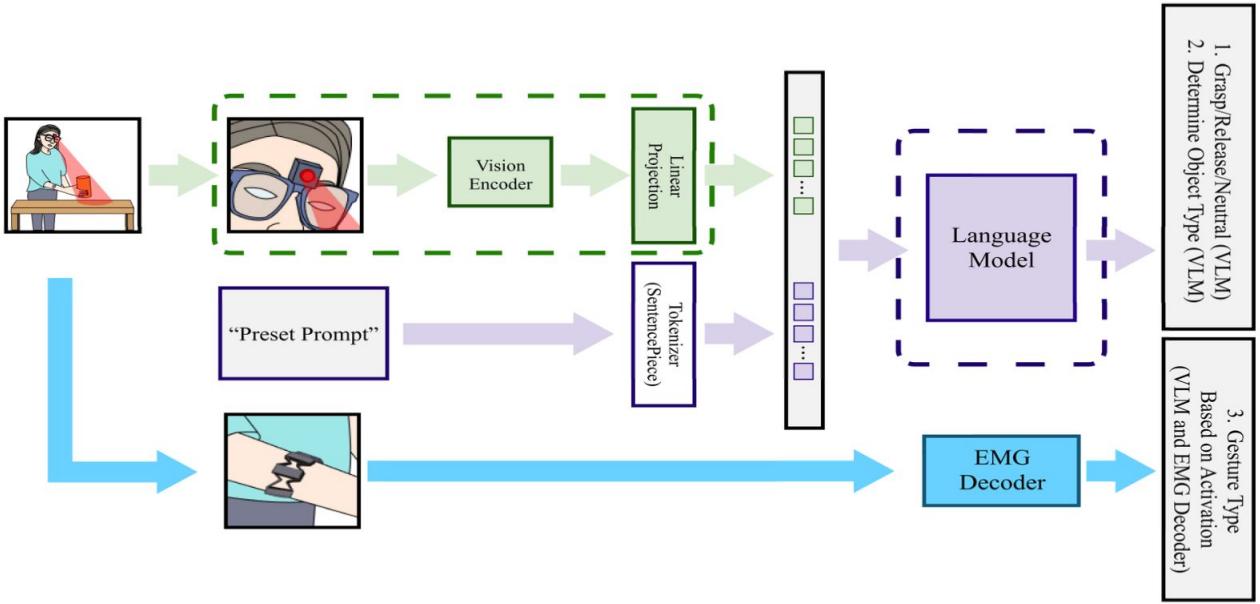


Fig. 1. Paligemma vision language model [13] and electromyography for user intention detection.

Reference architecture identified as the starting point for this project.

What is Found in the Existing Work?

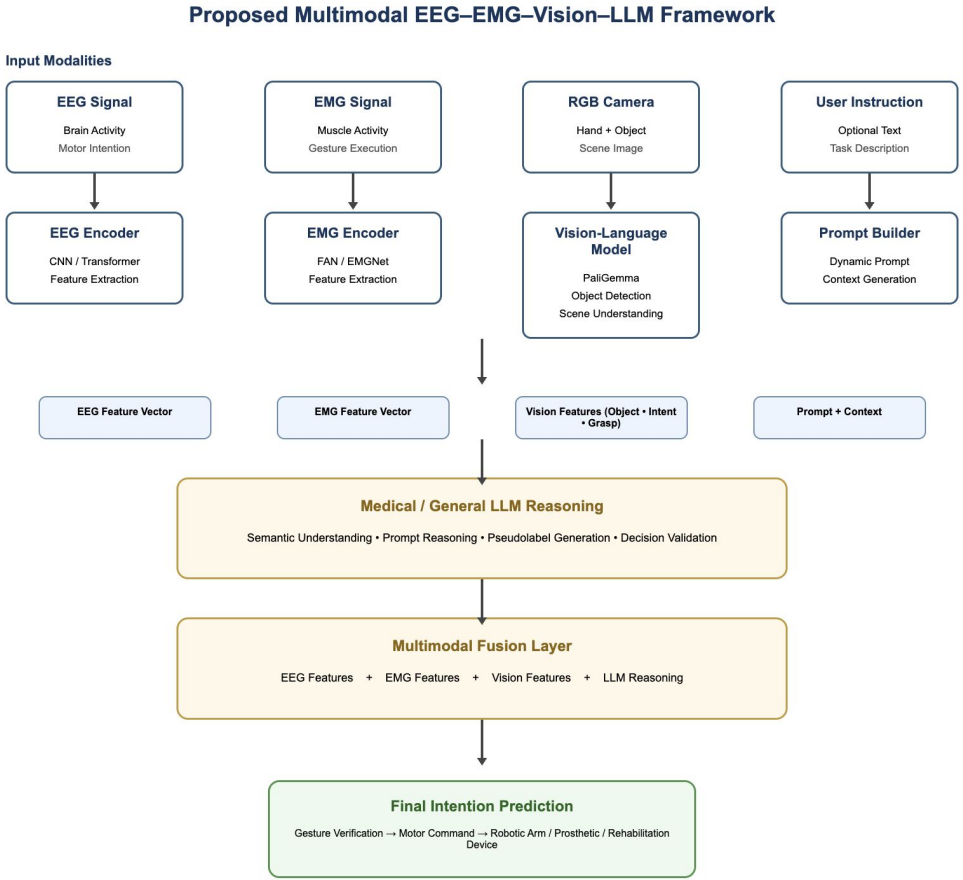
Component	Status
EMG dataset	✅ Available
EMG preprocessing	✅ Available
EMG decoder / FAN	✅ Available
EMG training notebooks	✅ Available
Vision pipeline	❌ Not implemented
PaliGemma implementation	❌ Not provided
Prompting pipeline	❌ Not provided
VLM–EMG fusion	❌ Not provided
Final multimodal decision logic	❌ Not provided

Reconstructing missing pipeline

We propose to reconstruct the missing branches using **appropriate publicly available models/data.**

Required component	Planned source	Model/Dataset
EEG	EEG dataset provided by my project guide	
EMG	Existing paper/repository dataset & decoder	
Image	Public hand/object vision dataset	**Hands23** dataset (45GB -> pick subset)
VLM	Pretrained vision-language model	**PaliGemma** like mentioned in paper)
Prompting	Task-specific prompts designed for intention & grasp reasoning	
LLM reasoning	Suitable lightweight instruction-following LLM	**Med-PaLM (medical) or Gemma3 (generic)**
Fusion	Proposed multimodal feature/decision fusion	
Final output	Intention + gesture verification	

Proposed Framework



Scope and Next Steps

Completed

- Studied **reference paper architecture**
- Inspected associated **GitHub** repository
- Verified availability of **EMG data and EMG processing code**
- Identified **missing VLM, prompting and fusion components**
- Investigated possible **public vision datasets**
- Defined initial **multimodal architecture**

Scope and Next Steps

Next Steps

1. Data

- Obtain EEG data from guide
- Select manageable subset of public vision data
- Establish compatible preprocessing

2. Models

- Reproduce EMG branch
- Implement EEG encoder
- Integrate pretrained VLM
- Evaluate suitable LLM

3. Integration

- Design prompts
- Generate VLM/LLM outputs
- Develop multimodal fusion
- Implement gesture verification

4. Evaluation

- Compare:
  - EMG only
  - EEG + EMG
  - EEG + EMG + Vision
  - Full multimodal framework

Literature Survey

Work	Modality	Key contribution	Limitation / gap
Dere et al. (2025)	EMG + Vision/VLM	VLM pseudolabels EMG gestures	No complete public VLM implementation
Rho et al. (2024)	Vision	Vision-based intention detection	Domain-specific curated data
NOIR / NOIR 2.0	EEG + Vision	Multimodal robotic control	Different task/application
EEG–EMG rehabilitation studies	EEG + EMG	Biosignal fusion	No VLM-based visual reasoning

Papers

1. **Target paper — VLM + EMG**
   M. D. Dere et al., “Lightweight Vision Language Model-Guided Gesture Recognition Based on Electromyography,” IEEE Sensors Journal, 2025.

2. **Vision-based intention detection / rehabilitation**
   E. Rho et al., “Multiple Hand Posture Rehabilitation System Using Vision-Based Intention Detection and Soft-Robotic Glove,” IEEE TII, 2024.

3. **Egocentric vision for user intention**
   D. Kim et al., “Eyes are faster than hands: A soft wearable robot learns user intention from the egocentric view,” Science Robotics, 2019.

4. **Why vision is important for wearable robots**
   L. Gionfrida et al., “Wearable robots for the real world need vision,” Science Robotics, 2024.

Novelty

Our Proposed Extension

① EEG + EMG Multimodal Biosignals
Integrate **EEG alongside EMG** to capture complementary information about motor intent.

② Unified Multimodal Representation
Bring together:

EEG + EMG + Vision + Language

rather than treating EMG–vision interaction as the primary pipeline.

③ Context-Aware Intent & Gesture Recognition
Use visual context and language understanding together with biosignals to distinguish:

- **What the user intends to do?**
- **What object is involved?**
- **How the action should be performed?**

④ Cross-Modal Validation / Decision Making
Use information from different modalities to **validate and disambiguate** the predicted gesture/intent, particularly in cases of EMG ambiguity or distribution shift.

Thank you!

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